

DP Computer Science: Topic A4 Practical Project — Course Recommender

Overview

This section provides a complete set of resources for the 3-day "Course Recommender" mini-project for Higher Level students. The project applies computational thinking to design a system that recommends a full, valid IB DP subject package (3 HL, 3 SL) based on a user's profile.

1. Teacher's Guide

Setup (Before Day 1)

Step	Description
Form Groups	Create groups of 2-3 students. Try to balance skill levels.
Introduce the Project	Explain the goal: to apply the computational thinking process to design a system that recommends a full, valid IB DP subject package (3 HL, 3 SL) based on a user's profile. Emphasise that this is a design project ; a working code implementation is an optional extension.
Distribute the Template	Share the "Student Project Brief & Template" with each group.

2. Project Planning

Day 1: Problem Specification, Decomposition, Abstraction

Aspect	Description
Focus	Guide students to define the more complex problem space. The key is to move from recommending a single group to a full package .
Common Pitfall	Students feel overwhelmed by the number of possible subject combinations.
Solution	Guide them towards the " archetype " approach (matching a user to a pre-defined career profile, e.g., "Future Engineer," "Future Diplomat"), which then has a valid subject package associated with it.

Key Questions for Students:

1. What are the different career archetypes you can identify?
2. What subjects are typically required for each archetype?
3. What skills or interests define each career path?

Day 2: Pattern Recognition, Algorithm Design

Aspect	Description
Focus	Designing a logical algorithm to match a user to the best archetype.

Aspect	Description
Common Pitfall	Students struggle to define the features for their archetypes.
Solution	Brainstorm as a class the kinds of skills or interests that define different career paths (e.g., Quantitative Reasoning, Scientific Inquiry, Creative Production, Historical Analysis). Guide them to create a multi-dimensional vector from these ideas.

Possible Algorithm Approaches:

Approach	Description
K-NN (k=1)	Match a user to the closest career archetype (nearest neighbour).
Decision Tree	Design a series of questions that forms a decision tree. Leaf nodes are the recommended subject packages.
Content-Based Filtering	Create "content" vectors for each subject package; match the user's interest vector to the package with the most similar content.

Day 3: Coding & Testing Plan

Aspect	Description
Focus	Thinking about implementation and robust verification.
Common Pitfall	Test cases are too simple.
Solution	Encourage students to create test profiles for users who are "in-between" two archetypes to see how the algorithm behaves. This tests the sensitivity of their distance calculations.

3. Project Brief for Students

The Problem

"A school wants to develop a system that helps students choose their IB DP subject package. The system should recommend a full, valid package of 6 subjects (3 HL, 3 SL) based on a student's interests, skills, and career aspirations."

Tasks

Task	Description
1. Problem Specification	Define the problem, constraints, goals, input, output, and evaluation criteria
2. Decomposition	Break down the problem into smaller, manageable parts
3. Abstraction	Identify the essential features and ignore irrelevant details
4. Pattern Recognition	Identify patterns in career paths and subject combinations

Task	Description
5. Algorithm Design	Design an algorithm to match users to subject packages
6. Testing Plan	Develop test cases to evaluate the algorithm

Example Archetypes

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CAREER ARCHETYPES

FUTURE ENGINEER

HL: Mathematics, Physics, Chemistry
SL: English, Language, Economics
Skills: Quantitative Reasoning, Scientific Inquiry

FUTURE DIPLOMAT

HL: History, Language A, Language B
SL: Mathematics, Economics, Science
Skills: Analysis, Communication, Cultural Awareness

FUTURE CREATIVE

HL: Art, English, Theatre
SL: Mathematics, Language B, Science
Skills: Creative Production, Expression

4. Possible Algorithm Approaches

K-NN (k=1) Approach

python

Example: Using k-nearest neighbours (k=1) to match a user to an archetype

Define archetypes with feature vectors

archetypes = {
 "Engineer": {
 "quantitative": 5,
 "scientific": 5,
 "creative": 1,
 "communication": 2,
 "subjects": ["Math HL", "Physics HL", "Chemistry HL", "English SL", "Language B SL", "Economics SL"]
 },
 "Diplomat": {
 "quantitative": 2,

```

    "scientific": 1,
    "creative": 2,
    "communication": 5,
    "subjects": ["History HL", "Language A HL", "Language B HL", "Math SL", "Economics SL", "Science SL"]
},
"Creative": {
    "quantitative": 1,
    "scientific": 1,
    "creative": 5,
    "communication": 4,
    "subjects": ["Art HL", "English HL", "Theatre HL", "Math SL", "Language B SL", "Science SL"]
}
}

```

```

def match_user(user_vector):
    """Match a user to the closest archetype."""
    closest = None
    min_distance = float('inf')

    for name, archetype in archetypes.items():
        distance = 0
        for feature in user_vector:
            distance += (user_vector[feature] - archetype[feature]) ** 2

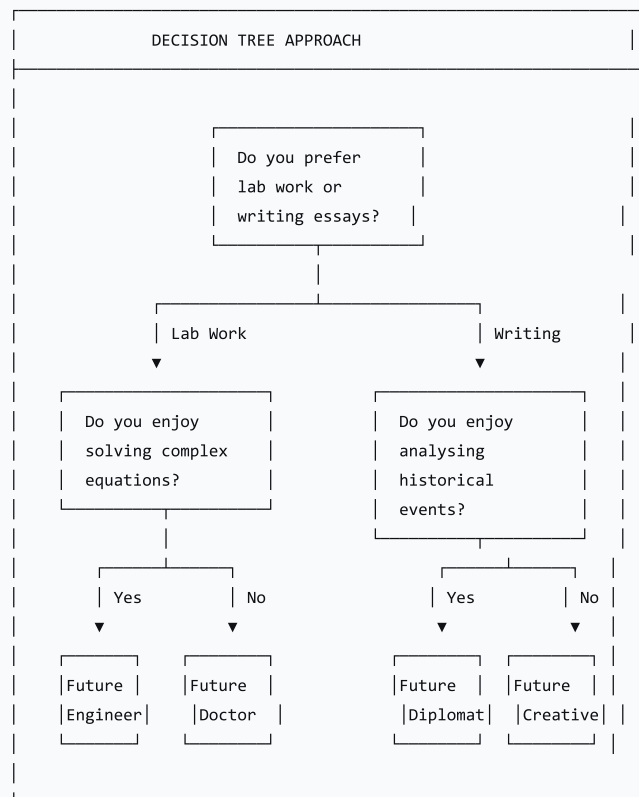
        if distance < min_distance:
            min_distance = distance
            closest = name

    return closest, archetypes[closest]["subjects"]

```

Decision Tree Approach

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5. Student Project Template

Section 1: Problem Specification

Element	Description
Problem Statement	
Constraints	
Goals	
Input Specifications	
Output Specifications	
Evaluation Criteria	

Section 2: Decomposition

Break down the problem into smaller parts:

1. **Data Collection** — What information is needed from the user?
2. **Archetype Definition** — What career archetypes exist?
3. **Matching Algorithm** — How will users be matched to archetypes?
4. **Subject Package Generation** — How will subject packages be generated?
5. **Validation** — How will the system be tested?

Section 3: Abstraction

Identify the essential features:

Feature	Description
Quantitative Skills	
Scientific Inquiry	
Creative Production	
Communication	
Analytical Skills	
Other	

Section 4: Pattern Recognition

Identify patterns in career paths:

Archetype	Key Skills	Required Subjects
Engineer	Quantitative, Scientific	Math HL, Physics HL, Chemistry HL
Diplomat	Analysis, Communication	History HL, Language A HL, Language B HL

Archetype	Key Skills	Required Subjects
Creative	Creative, Expression	Art HL, English HL, Theatre HL

Section 5: Algorithm Design

Design an algorithm to match users to subject packages:

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Algorithm: Subject Recommender

1. Input: User profile (skills, interests, aspirations)
2. For each archetype:
  a. Calculate similarity to user profile
3. Select archetype with highest similarity
4. Generate recommended subject package
5. Output: 6 subjects (3 HL, 3 SL)
```

Section 6: Testing Plan

Test Case	Input	Expected Output	Justification
1	User with high quantitative skills	Future Engineer	Strong match with quantitative archetype
2	User with high communication skills	Future Diplomat	Strong match with diplomatic archetype
3	User with in-between profile	Closest match	Tests sensitivity of distance calculation

6. Extension Activities

Activity	Description
K-NN Implementation	Implement the algorithm using scikit-learn's KNeighborsClassifier
RNN Exploration	Explore TensorFlow Basic RNN Tutorial for text generation
Public Datasets	Use Kaggle or UCI Machine Learning Repository for additional data

7. Project Resources

Resource	Description
Student Project Template	Template for students to complete
Project Exemplar	Model answer/example project
K-NN Documentation	Scikit-learn KNeighborsClassifier Documentation
RNN Tutorial	TensorFlow Basic RNN Tutorial

Resource	Description
Datasets	Kaggle, UCI Machine Learning Repository, Google Dataset Search

8. Summary: Key Takeaways

Concept	Key Point
Problem Specification	Define the problem clearly
Decomposition	Break the problem into smaller parts
Abstraction	Focus on essential features
Pattern Recognition	Identify patterns in career paths
Algorithm Design	Create a logical matching algorithm
Testing	Develop robust test cases

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KEY REMINDERS

✔

Design project, not just coding

✔

Use archetypes to manage complexity

✔

Consider multiple algorithm approaches

✔

Create robust test cases

✔

Documentation and design are assessed

"The Course Recommender project applies computational thinking to a real-world problem—helping students make informed decisions about their IB DP subject choices."